

Osman Akar

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EDUCATION

PhD in Applied Mathematics at University of California, Los Angeles <i>Thesis: Novel Applications of Neural Networks in Physics-Based Simulations</i>	2019 - 2024
Masters in Mathematics at University of California, Los Angeles <i>Departmental Scholars Program (simultaneous BS and MA degrees awarded)</i>	2016 - 2019
BS in Mathematics of Computation at University of California, Los Angeles <i>Math Undergraduate Merit Scholar, Graduated Cum Laude</i>	2015 - 2019

PROJECTS & RESEARCH

Research Intern at Epic Games Machine Learning based Shallow Signed Distance Functions	Los Angeles, CA Jun 2022 - December 2024
- Developed a novel learning-based method to compute signed distance functions (SDF) for human avatars with dynamically changing body shapes due to joint-based skinning. Implemented this method as a custom asset within Unreal Engine, leveraging its capabilities for cloth simulation. Optimized cloth interaction memory usage in character animations within Unreal Engine, achieving a 10x reduction in memory cost. - Created a comprehensive training dataset pipeline by: (1) Generating various human avatar poses using Autodesk Maya and exporting them in a C++-friendly format, (2) Computing raw signed distance data for the posed avatars using C++, (3) Processing raw data into a trainable format with Python. - Trained machine learning models using TensorFlow to enhance performance and accuracy of SDF-based simulations.	
Graduate Researcher at Teran Lab A Deep Conjugate Direction Method (DCDM) for Iteratively Solving Linear Systems	Los Angeles, CA Feb 2021 - May 2022
- Created CNN-based linear system solver, a modified version of the classical Conjugate Gradients (CG) method, for the systems coming from computational fluid dynamics applications (e.g. incompressible flows). - Designed the neural network architecture and its custom loss function to perform unsupervised learning to speed up the solver. DCDM is 3.2 times faster than classical methods for linear systems with over 56M degrees of freedom. - Developed a novel method for creating the dataset that does not require any simulation input, as opposed to our competitors. This reduced the dataset size, the generation and the training time by over 2x. - The work is accepted and presented at ICML 2023 at Hawaii. See https://icml.cc/virtual/2023/poster/24899	

TEACHING

- **Instructor** of *Introduction to Probability* class of 40 students at 23 Spring.
- **Teaching Assistant** of Probability, Algorithms, Applied Ordinary Differential Equations (Graduate Level), Linear and Nonlinear Systems of Differential Equations, Analysis, and Calculus classes since 19 Fall.
- **Lead of the Olympiads Instructor Group** in [Olga Radko Math Circle \(ORMC\)](#) for 21-22 and 22-23 school years.

HONORS AND AWARDS

- O'Neill Travel Award by UCLA Mathematics Department, 2024
- Putnam Performance Prize by UCLA Mathematics Department, 2019
- Honorable Mention at [William Lowell Putnam Mathematical Competition](#), 2018. Ranked 77 among 4623 students.
- Richard F. Arens Putnam Scholars Undergraduate Award by UCLA Mathematics Department, 2016
- Sole recipient of [UCLA Math Undergraduate Merit Scholarship](#) in class of '19. Since its foundation at 2010, there have been only 8 recipients of this prestigious scholarship all over the world.
- **Gold Medal** at International Mathematical Olympiad (**IMO**), 2014, South Africa
- Silver Medal at International Mathematical Olympiad (**IMO**), 2013, Colombia

PROGRAMMING & SKILLS

Skills: MACHINE LEARNING, COMPUTATIONAL AND APPLIED MATHEMATICS, COMPUTATIONAL LINEAR ALGEBRA, CONVOLUTIONAL NEURAL NETWORKS, COMPUTER GRAPHICS, NUMERICAL METHODS, ALGORITHMS

Languages: C++, PYTHON, TENSORFLOW, MATLAB, L^AT_EX, MAYA, HOUDINI